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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 38: Concept of Circular Economy

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Contents

- Economy & current practice
- Drawbacks of linear model of economy
- Concept of Circular Economy (CE) & its definition
- Ideas behind the concept of CE
- Model for CE
- Biological & technological cycles
- Benefits of CE

What is economy?

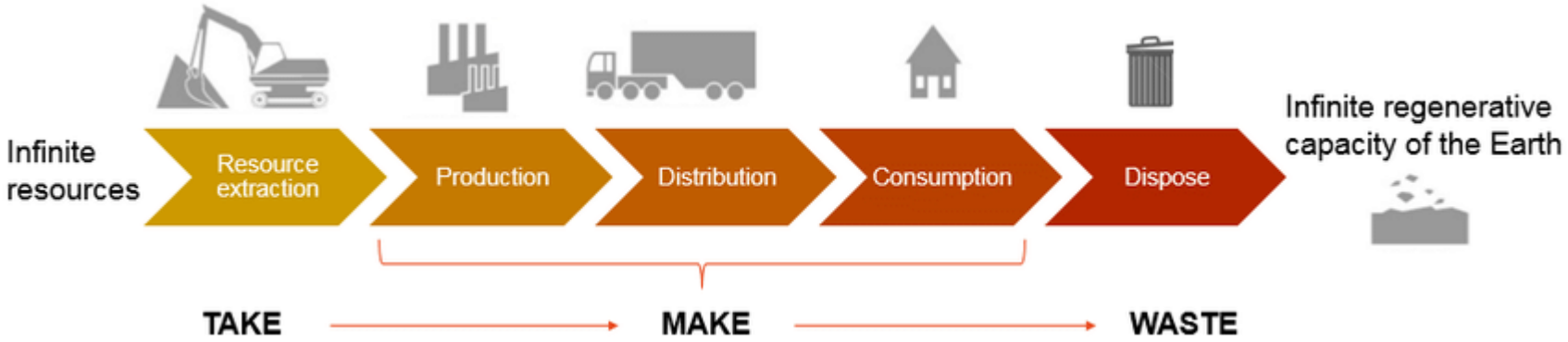
- Economy is concerned with the production, distribution, and consumption of goods and services.
- It studies how individuals, businesses, governments, and nations make choices about how to allocate resources.

- Economy is generally considered on national level but can be considered at local, regional and also global level.
- In the era of global village, all economies are closely interlinked with each other.
- General parameter of assessing the economy is GDP (Gross Domestic production).

The operation of money supply, commercial activities/services and industry are important components of modern economy..

Source: Oxford dictionary, Investopedia.com

Linear model: Current practice of economy



- Linear economic model is based on a 'take-make-consume-throw away' pattern.
- This model based on ‘higher the consumption-bigger the economy’ and related activities.

Image Source: Wautelet, Thibaut. 2018

Drawbacks of linear economy



10-15% of building material wasted during construction



60% of European offices are not used even in working hours



20-40% of energy in existing buildings can be profitably conserved



54% of demolition materials landfilled, while some countries only landfill **6%**

Drawbacks of linear economy

- 2-5 % of global GDP is lost due to congestion
- Roads are capturing more and more land in urban spaces
- Air-quality is worsening around the globe
- One third of food produced globally - worth USD 1 trillion - is wasted each year, yet more than 10% of the world's population goes hungry

Source: UN Habitat, 2013, Ellen MacArthur Foundation portal for Circular Economy

Solution?

- Is prevailing situation sustainable? No, then what should be the approach?

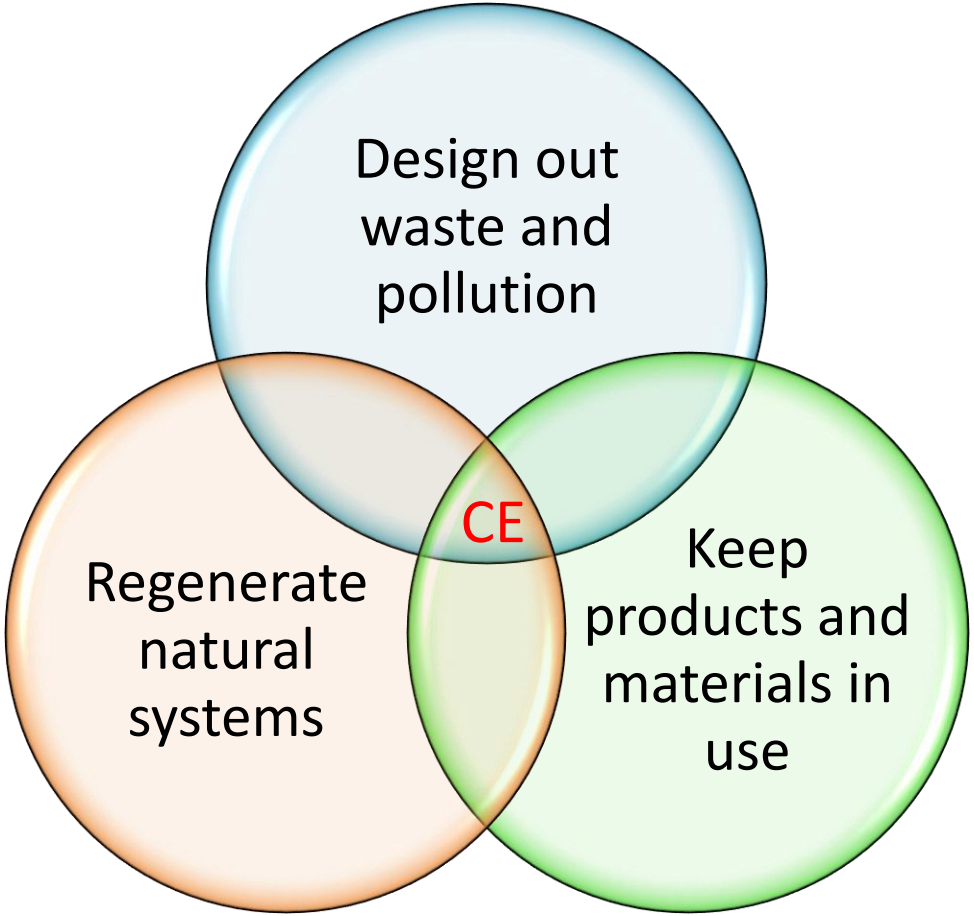
- Conceptual Solution
  - Minimize the consumption and wastage
  - Optimum utilization of resources
  - Regenerate and reuse the products or services
  - Return the resources to the environment

- Model for following these concepts into practice can be termed as ‘Circular Economy’ (CE)

Circular economy: Definition

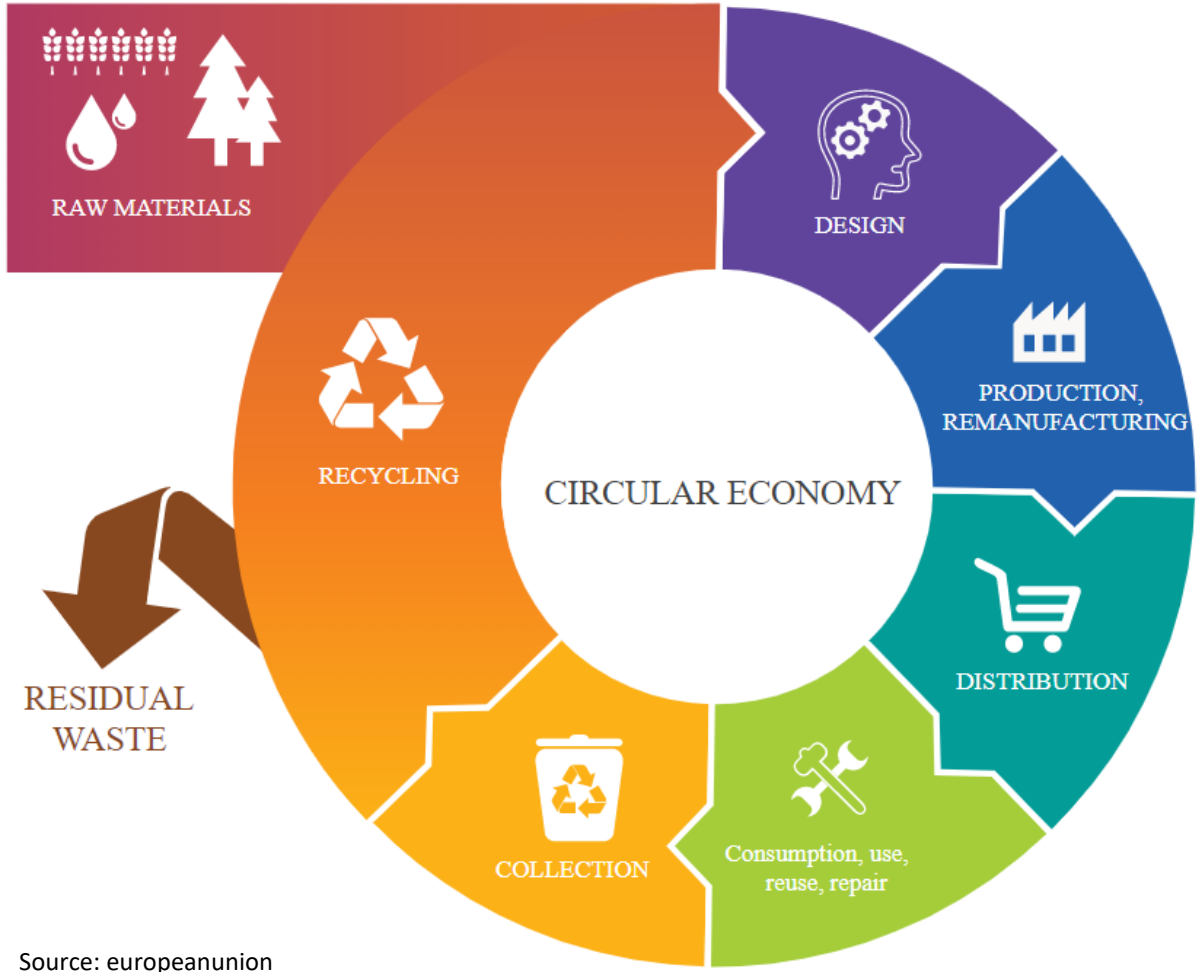
- A circular economy is a systemic approach to economic development designed to benefit businesses, society, and the environment

- A production and consumption model which involves reusing, repairing, refurbishing and recycling existing materials and products to keep materials within the economy wherever possible.



Principles of Circular Economy

Circular economy: Conceptual Diagram



Source: europeanunion

Circular Economy : Features

- An economy that is restorative and regenerative by design
- In a circular economy, economic activity builds and rebuilds overall system health
- Implies that waste will itself become a resource, consequently minimizing the actual amount of waste
- The concept recognizes the importance of the economy needing to work effectively at all scales – for big and small businesses, for organizations and individuals, globally and local

Linear economy Vs. Circular Economy

Take-make-dispose
Vs.
Make-Use-Return
Or
Make-consume-enrich

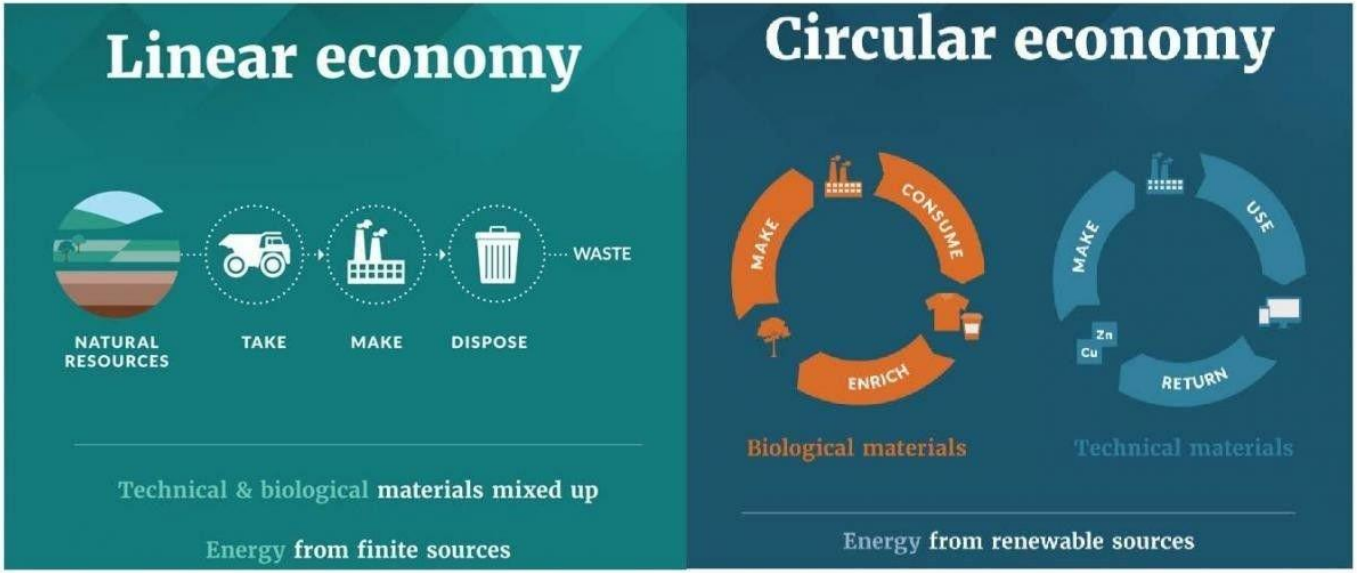


Image Source: plasticocene.org

Ideas behind CE: Performance economy

Performance economy based on:

- Product-life extension
- Long-life goods
- Reconditioning activities
- Waste prevention

Selling service Vs. product

Many products acquired by us can be rented for time being rather than permanently owning them

- Insists on the importance of selling services rather than products

- Idea referred to as the ‘functional service economy’, now more widely subsumed into the notion of ‘performance economy’.

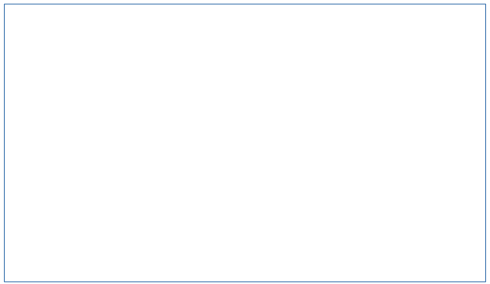
Ideas behind CE: Cradle to Cradle 1/2

- Design philosophy considers all material involved in industrial and commercial processes to be nutrients
- Framework focuses on design for effectiveness in terms of products with positive impact and reducing the negative impacts of commerce through efficiency
- Perceives the safe and productive processes of nature's 'biological metabolism' as a model for developing a 'technical metabolism' flow of industrial materials
- Product components can be designed for continuous recovery and reutilization

Ideas behind CE: Cradle to Cradle 2/2

Based on the following:

- Eliminate the concept of waste
- Power with renewable energy
- Respect human & natural systems
- Manage water use to maximize quality
- Promote healthy ecosystems
- Respect positive local impacts
- Guide operations and stakeholder relationships using social responsibility



Ideas behind CE: Biomimicry 1/2

Janine Benyus, author of Biomimicry: Innovation Inspired by Nature, defines approach as:

A new discipline that studies nature's best ideas and then imitates these designs and processes to solve human problems. Studying a leaf to invent a better solar cell is an example.



Image: Suitable aerodynamic design of high-speed rail inspired by nature

Source: ecomatcher.com

Ideas behind CE: Biomimicry 2/2

Nature works as model, measure & mentor for process & system.

Model	Measure	Mentor
Study nature's models and emulate these forms, process, systems, and strategies to solve human problems	Use an ecological standard to judge the sustainability of our innovations	View and value nature not based on what we can extract from the natural world, but what we can learn from it

Ideas behind CE: Industrial Ecology

- Study of material and energy flows through industrial systems
- Aims at creating closed-loop processes in which waste serves as an input, thus eliminating the notion of an undesirable by-product.
- Adopts a systemic point of view, designing production processes in accordance with local ecological constraints whilst looking at their global impact from the outset
- Industrial system perform as close to living systems as possible
- Principles can also be applied in the services sector

Ideas behind CE: Natural Capital

- “Natural capital" refers to the world’s stocks of natural assets including soil, air, water and all living things.

- Book “Natural Capitalism: Creating the Next Industrial Revolution” describe a global economy in which business and environmental interests overlap, recognizing the interdependencies that exist between the production and use of human-made capital and flows of natural capital.

Ideas behind CE: Principles for Natural Capital

- Radically increase the productivity of natural resources
- Adopt nature's designs where every output is either returned harmlessly to the ecosystem as a nutrient or becomes an input for another manufacturing process
- Move to a "service-and-flow" business model - Providing value as a continuous flow of services rather than the traditional sale-of-goods model
- Reinvest in natural capital - As human needs expand and pressures on natural capital mount, the need to restore and regenerate natural resources increases

Ideas behind CE: Blue economy & Regenerative design

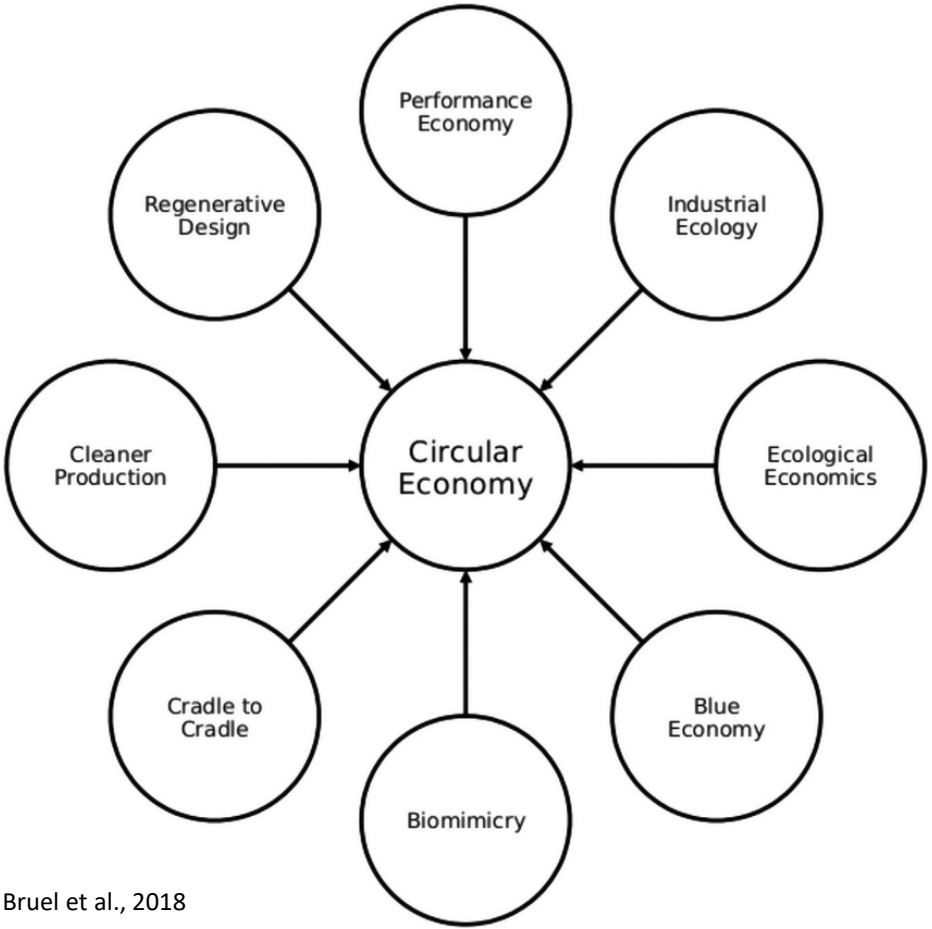
- Based on 21 founding principles & ocean driven economy.
- Insists on solutions being determined by their local environment and physical/ecological characteristics
- Putting the emphasis on gravity as the primary source of energy.

- Concept of regenerative design formulated for agriculture sector.
- Secondary products/waste used in other process or works as nutrients for natural system.
- Applicable in many sectors.

Idea behind CE: User Vs. Consumer & sharing economy

- Raises questions about the necessity of owning products in the way that we traditionally do.
- Access to the service a product provides that is important, rather than the product itself.
- Understanding this shift in mindset lays the groundwork to many of the practicalities of shifting our economy from linear to circular.

Ex.- We need to travel or own the vehicle? Shared mobility can be the replacement of owning the car.



Source: A. Bruel et al., 2018

Concept of Circular economy is based on multiple ideas & theories.

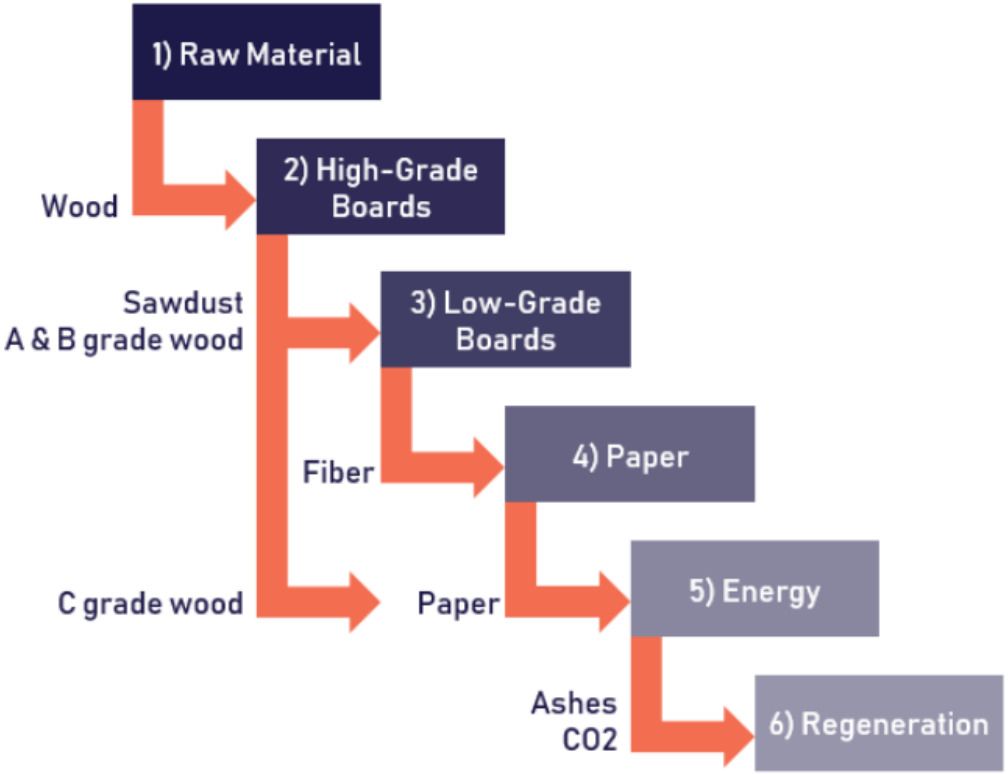
Cascades

- Process of putting used material & components for different re-uses.

- It is a repetitive cycle until product returns to the nature as nutrients

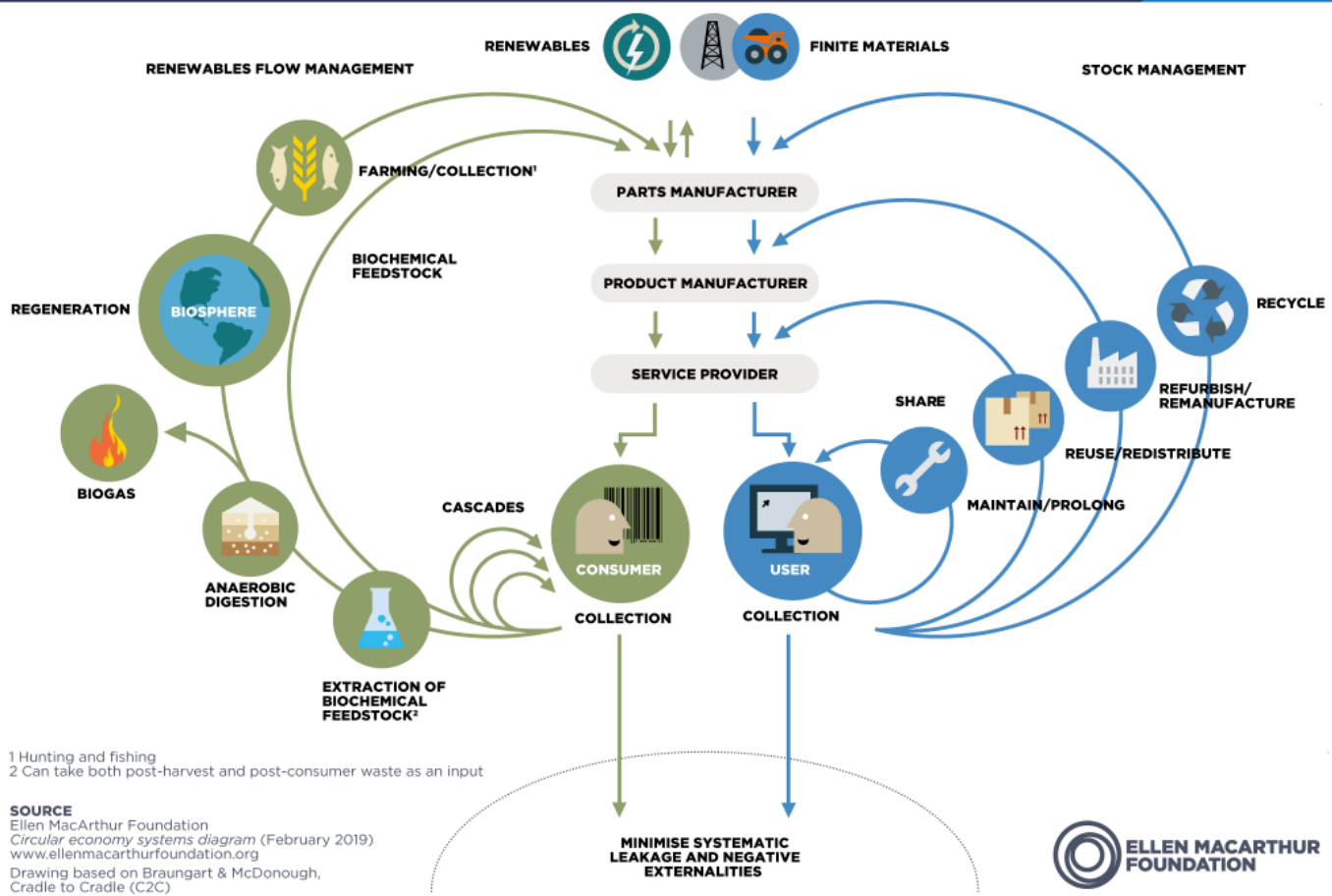
Ex.- Used paper recycled and reused as paper again or other products like packing material, used in construction as filling material and then retuned to nature (easily decomposable)

Example: Cascades Wood

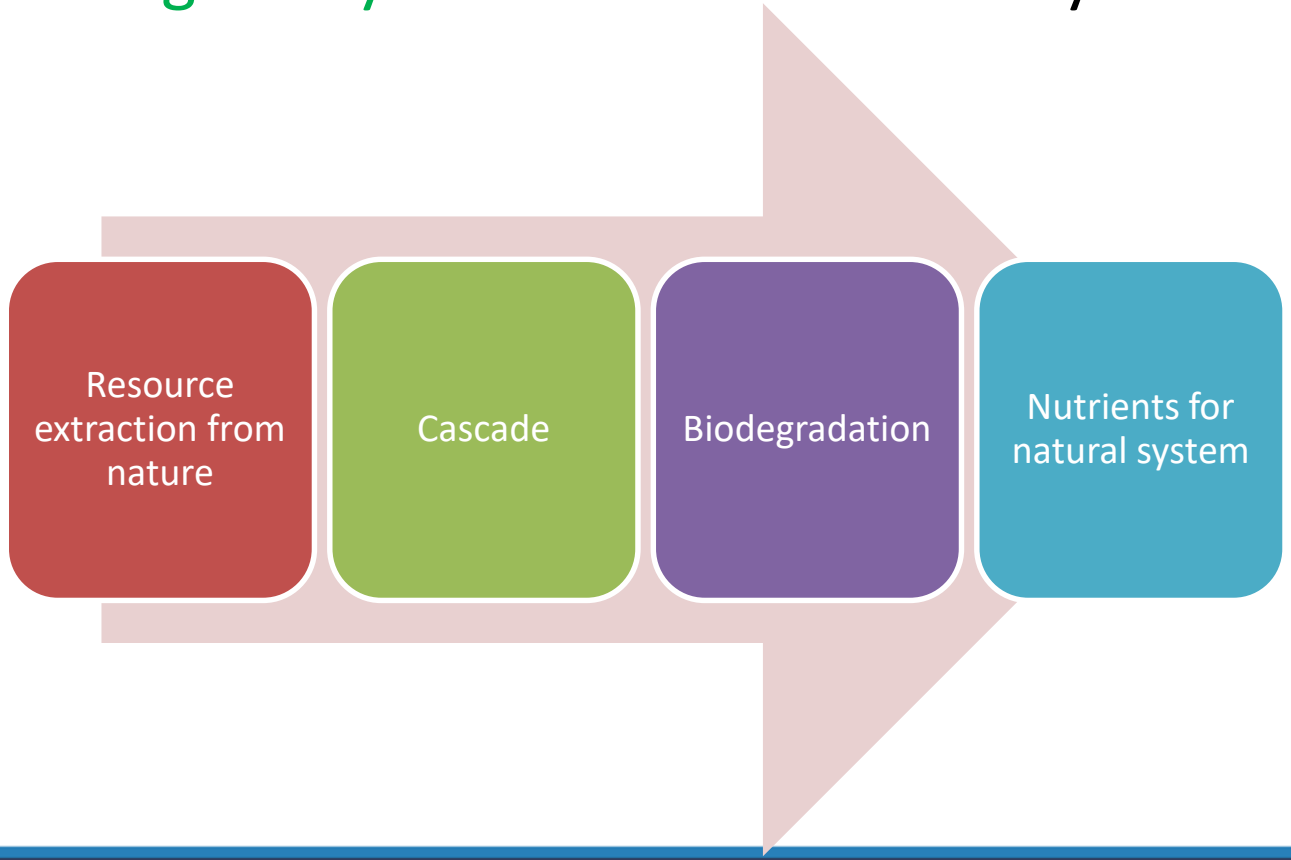


Source: community.materialtrader.com

Butterfly diagram: Circular economy Model



Biological Cycle of Circular economy

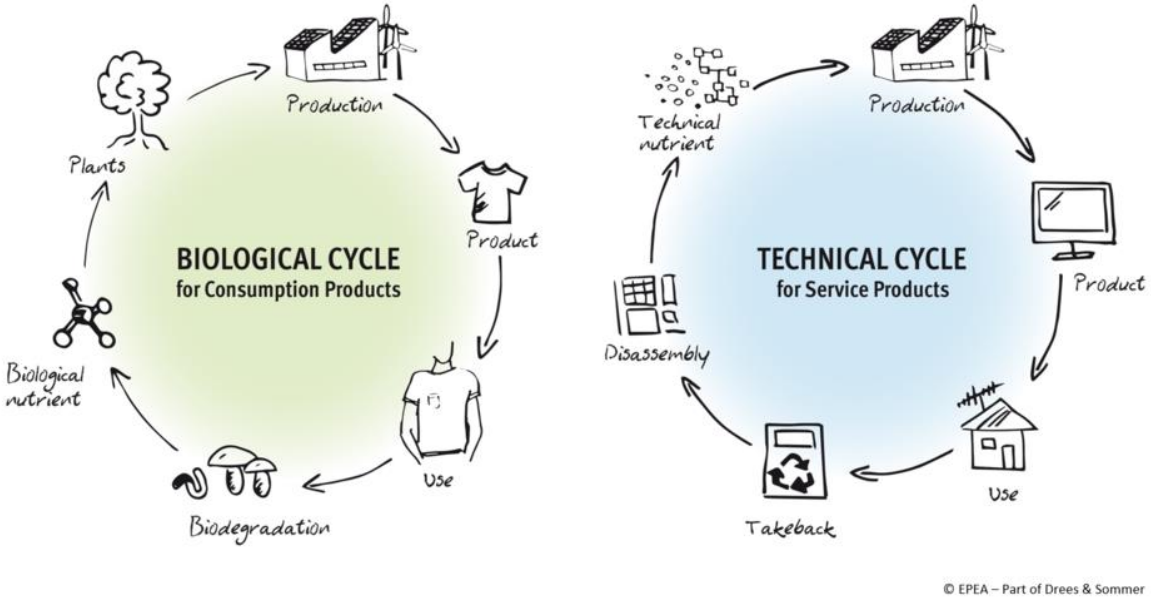


- Represented as green cycles on the left side of the butterfly diagram

Technological cycle of Circular economy

- Cycle/loop represented in blue on the right-hand side of butterfly diagram
- Consists technological material which cannot re-enter the environment
- These materials, such as metals, plastics, and synthetic chemicals, must continuously recycle through the system so that their value can be captured and recaptured

Biological & Technical Cycle: Simplified



Source: epea.com

Maintain-prolong & share	Refurbish & remanufacture	Recycle
Keeping product & material in use by prolonging the life span Design with durability & proper maintenance/repair Shared among users which will reduce consumption of material/product.	Process of restoring the value of product. Refurbish process doesn't disassemble the product and does cosmetic make over of product. Under remanufacture process, whole product is disassembled into basic units and necessary parts are replaced before redistribute	Reducing the product all the way back to its basic material Most or at least few parts to be remade into new products. Compare to reuse or refurbish it has low value due to: • Extra labor & energy involved • Inevitable material loss • Cost

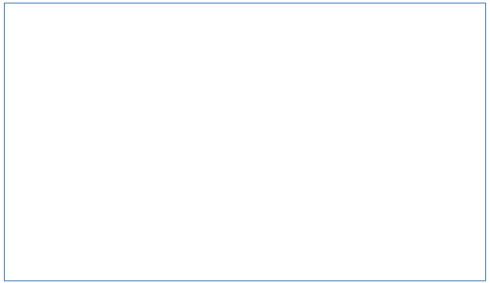
Loops of Technological cycle

Economic Benefits of Circular Economy

- Job creation potential
  - Labor intensive recycling
  - Higher skilled job in manufacturing compared to low skilled job
  - New serviced based economy
- Innovation
  - For recycling, remanufacturing & resource savvy production
- Material cost saving
- Economic growth
  - Through combination of increased revenues from circular activities & more productive utilization of inputs.

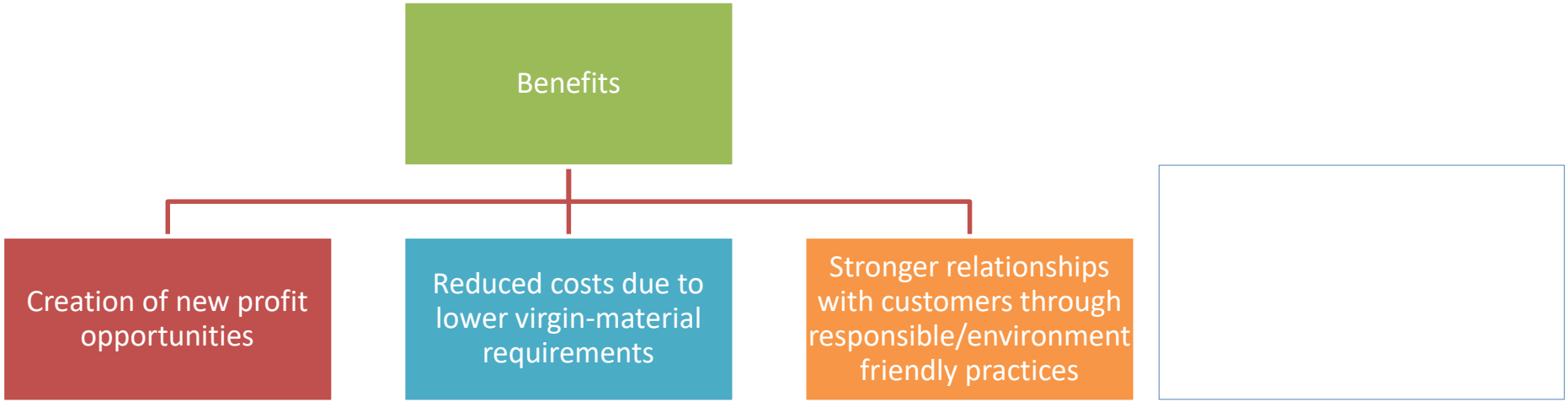
Environmental Benefits of Circular Economy 1/2

- Circular economy represents a powerful contribution to achieving global climate targets by:
  - Designing out waste and pollution
  - keeping products and materials in use
  - Regenerating rather than degrading natural systems



Benefits of CE for Industries

- Businesses would benefit significantly by shifting their operations in line with the principles of the circular economy.



Way for Linear to Circular Economy: System Thinking

- Business models, design, legislation, planning, practices, manufacturing, and more, all currently have undesirable qualities from a circular perspective.
- Cannot change just one element of the existing system and expect the change we need.
- So, need to look up on the holistic approach rather than micro interventions

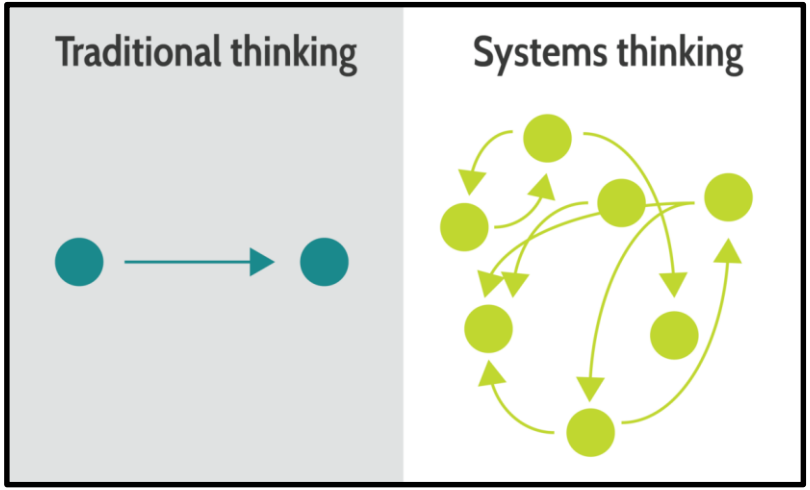


Image source: altus.education

Ancient environment friendly practices in India

- As per Indian culture, domestic animals are part of family not just a resource.
- Indian farming always used organic fertilizers (made of dung & organic waste) before introduction of chemical fertilizers.
- Concept of sacredness in trees placed moral values for environment conservation.
- All resources like rivers, forests or earth treated like living being which now adopted in modern concepts like system metabolism
- Affection for ancestral property also a common feature in rural India which ensures protection of wells, resources and land.

Direct-indirect imposition of religious-moral & social value about nature creates permanent in-built value system and promote conservation behavior.

Indian philosophy schools about elements of the environment : Vedic Darhsan

माता भूमि पुत्रोहं पृथिव्या”

(Mata Bhumi putroham prithivyah)

Meaning: – “Earth is my mother; I am her son”

Do not harm the environment, do not harm the water and the flora, earth is my mother, I am her son, may the waters remain fresh, do not harm the waters—-. Tranquility be to the atmosphere, to the waters, to the crops and vegetation.

- Vedas consists hymes in which prayers to Indra (God) ‘not to separate trees from the forests and the sons from their fathers’ (Rig-veda, 8/1/13)
- The Vṛkṣāyurveda says that planting a tree is equally beneficial as having ten offspring.

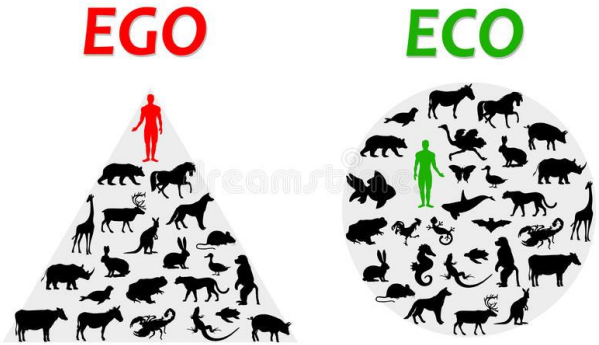
Source: Rajib Sarmah, 2015

Indian philosophy schools about environmental conservation :

Buddhism

According to Buddhism

- Environment and its components are not just to serve humans.
- Human beings are only its part and should not show their supremacy over the different biotic and abiotic components of nature.
- Several of the literary works establishes Buddhism as an eco-centric philosophy.
- According to Kutadanta, it is the responsibility of the ruler to protect flora and fauna of his/her state.



Indian philosophy schools about environment conservation : Rabindranath Tagore & Mahatma Gandhi

"The highest education is that which does not merely give us information but makes our life in harmony with all existence".
-Rabindranath Tagore

- Every major Indian philosophical school of thought emphasized on conservation and protection of nature or the environment and to live in harmony with it.
- Mahatma Gandhi propagated the concept of 'Gram Swarajya' which is based on self-reliance & environment friendly development of rural economy.

Conclusions

- Current economic model i.e. linear economic model is leading us to unsustainable situation.
- Circular economy can be the replacement of current linear model to promote sustainable practices.
- Concept of CE is based on various older / ancient time-tested concepts.
- Environmental, economic and industrial benefits are ensured through application of CE.
- A holistic approach need to be followed for implementation of CE.

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Thank You